Industrial Path Solutions

IPS IMMA Virtual Driver R&D

Industrial
Path
Solutions

IPS IMMA for Vehicle Ergonomics

IPS IMMA

Fast and efficient algorithms for easy **evaluation of assembly ergonomics** that considers human diversity using a realistic biomechanical model.

RESEARCH PROJECTS WITH VEHICLE ERGONOMICS FOCUS (IN SWEDEN)

- VD (Virtual Driver) 1-year, 2015-2016
- VDE (Virtual Driver Ergonomics) 2-year, 2017-2019
- ADOPTIVE (Automated Design and Optimization of Vehicle Ergonomics) 3-year, 2020/2021-2024

Progression from development of core seated posture functionality to evaluation and refinement of developed functionality as well as identification of future research and development topics

IPS IMMA Virtual Driver R&D

R&D results

Core functionality

- Attachment points and seated strategies
- Vision analyzer

Proof of concept functionality

- Statistical posture prediction models
- Forward simulation engine
- Manikins with Statistical Body Shape Models
- Simulation Based Multi Objective Simulation
- Reach envelope evaluation
- Vision analysis
- Dynamic simulation of ingress
- User-centred simulation process

Additional R&D topics

Core functionality

- Body shape analysis and anthropometrics
- Hip joint centre and pelvis rotation
- H-point to mid-hip prediction model
- Vision constraints affecting postures (direct and indirect)
- Semi-automatic simulation setup functionality
- Scenario operations simulation
- Ergonomics evaluation
    - Posture comfort evaluation
    - Reach comfort evaluation
    - Force/muscle evaluation
    - Vision evaluation (visualisation and export)
    - Collision/clearance evaluation
- Optimisation procedure for vehicle ergonomics
(static posture prediction and scenario operations simulation)

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